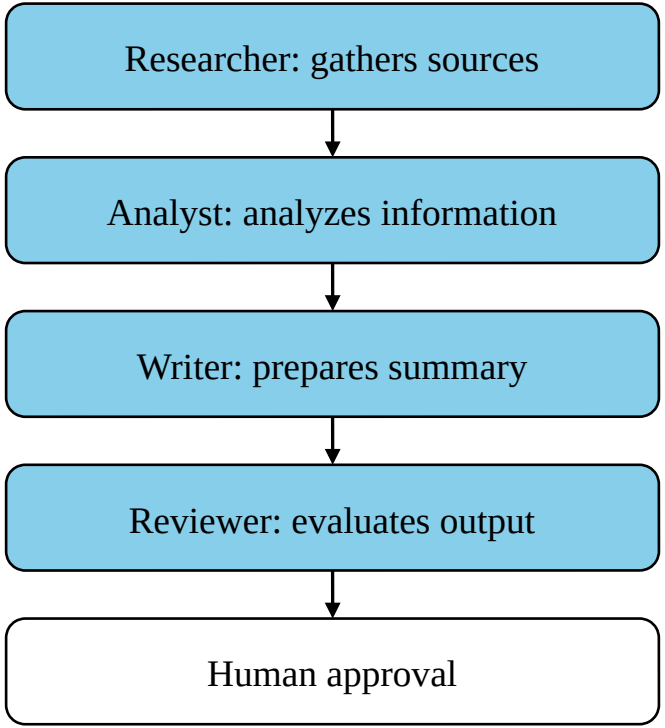
Autonomous AI Agent Workflow Orchestration Platform

Team Athena
Yogesh Aravind Karthikeya Kancherla

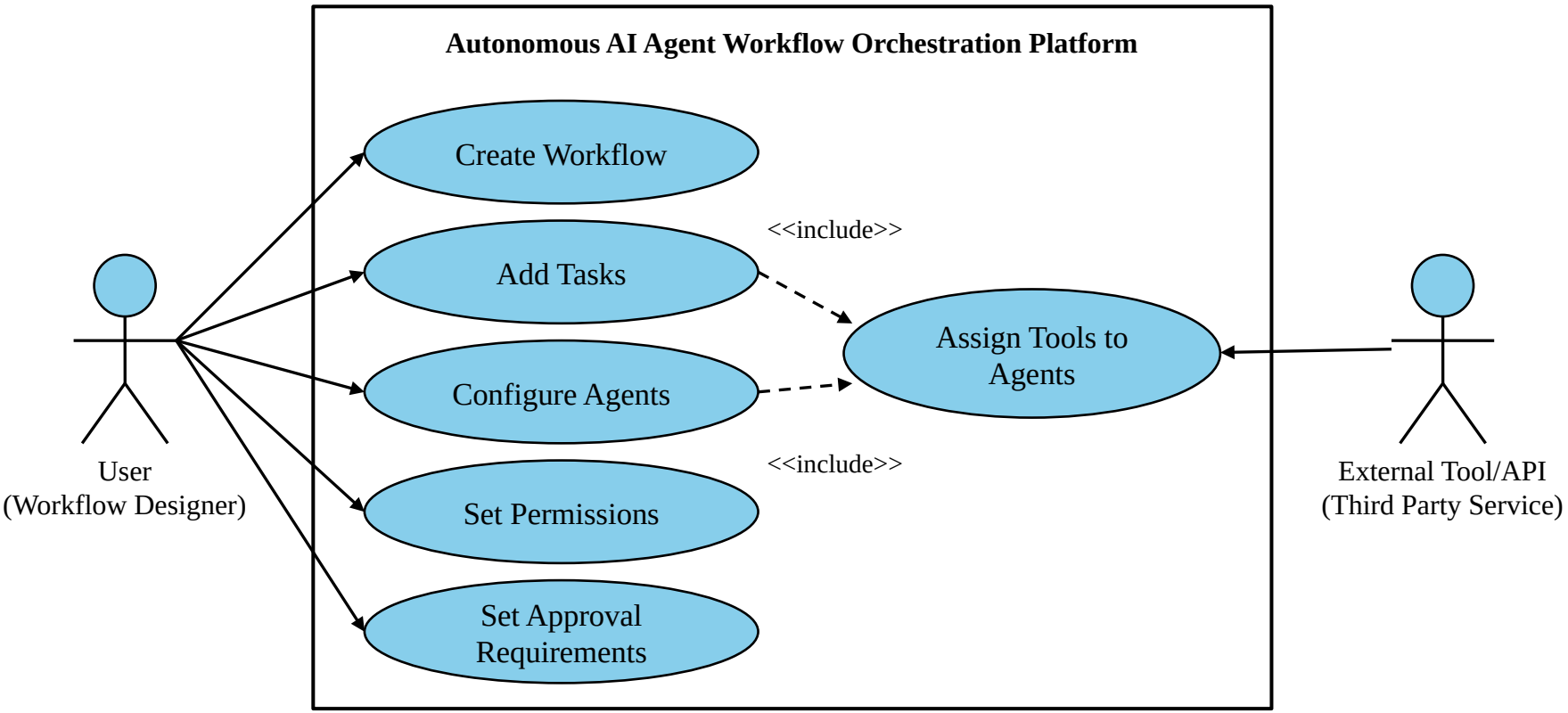
Project Status Presentation | October 28

Project Overview

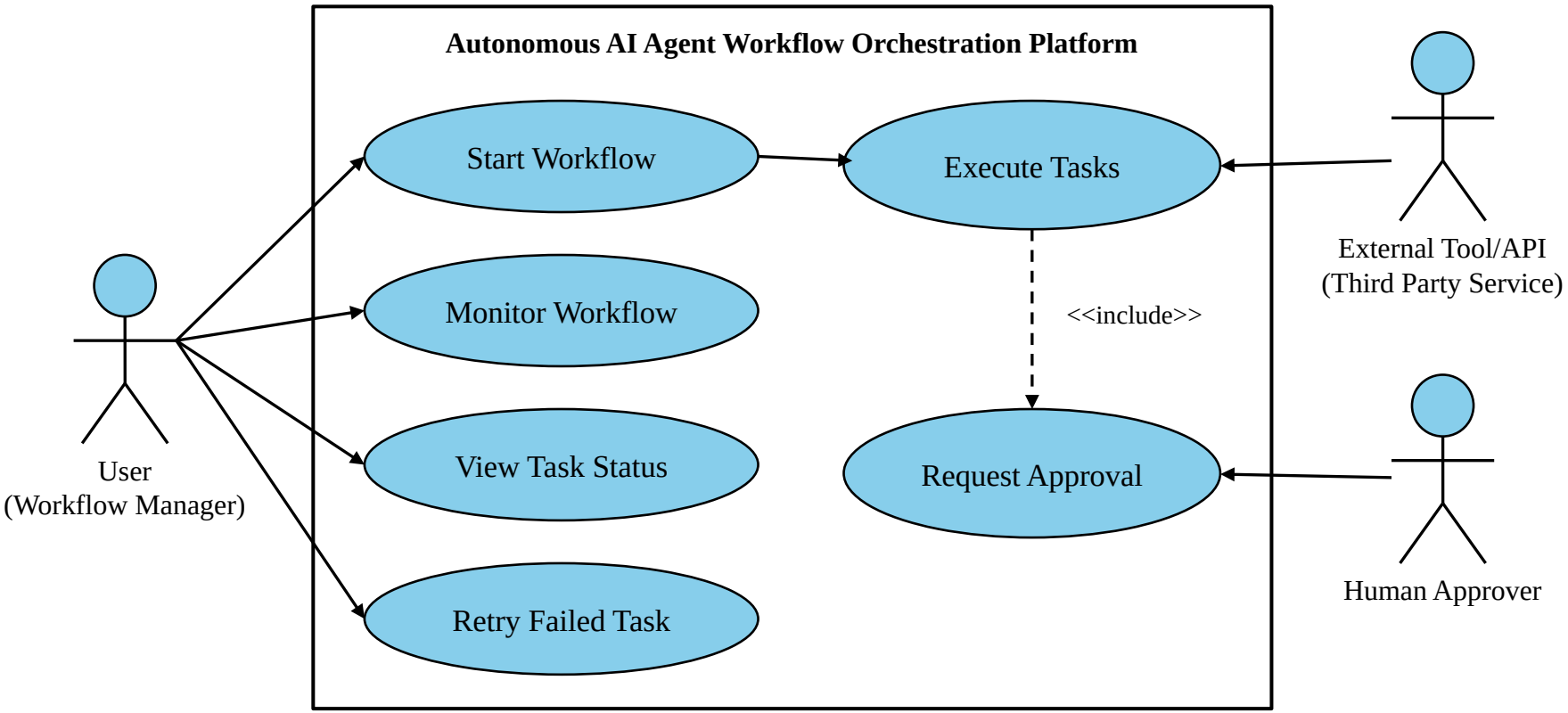
- Create and manage multiple AI agents that collaborate on complex, multi-step tasks
- User gives a high-level goal; an AI orchestrator breaks it into smaller tasks and assigns them to specialized agents
- Sequential or parallel execution with task dependency management
- Workflow visualization, agent configuration, access control, real-time monitoring and error handling
- High-impact actions pause for human approval



Use Case Diagram 1: Create and Configure Workflow



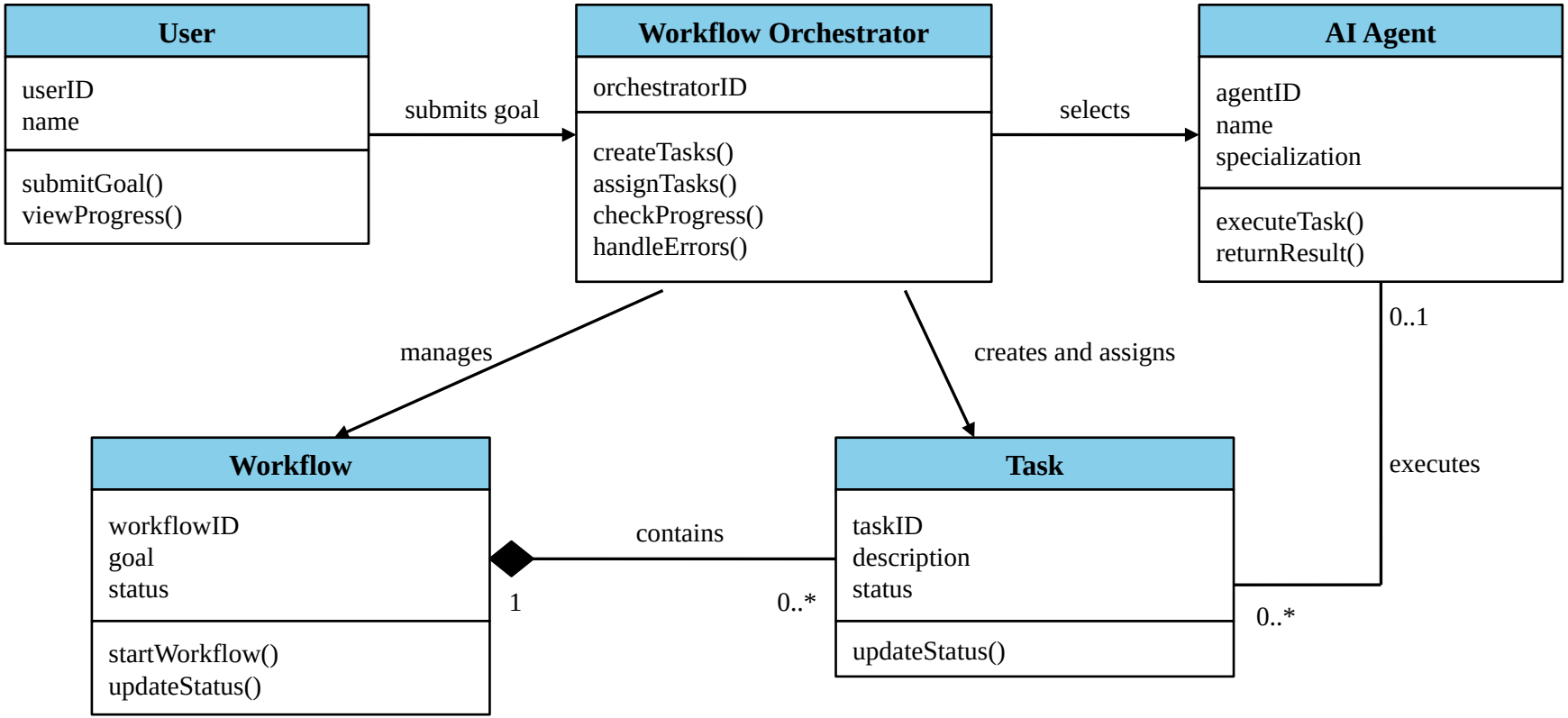
Use Case Diagram 2: Execute and Monitor Workflow



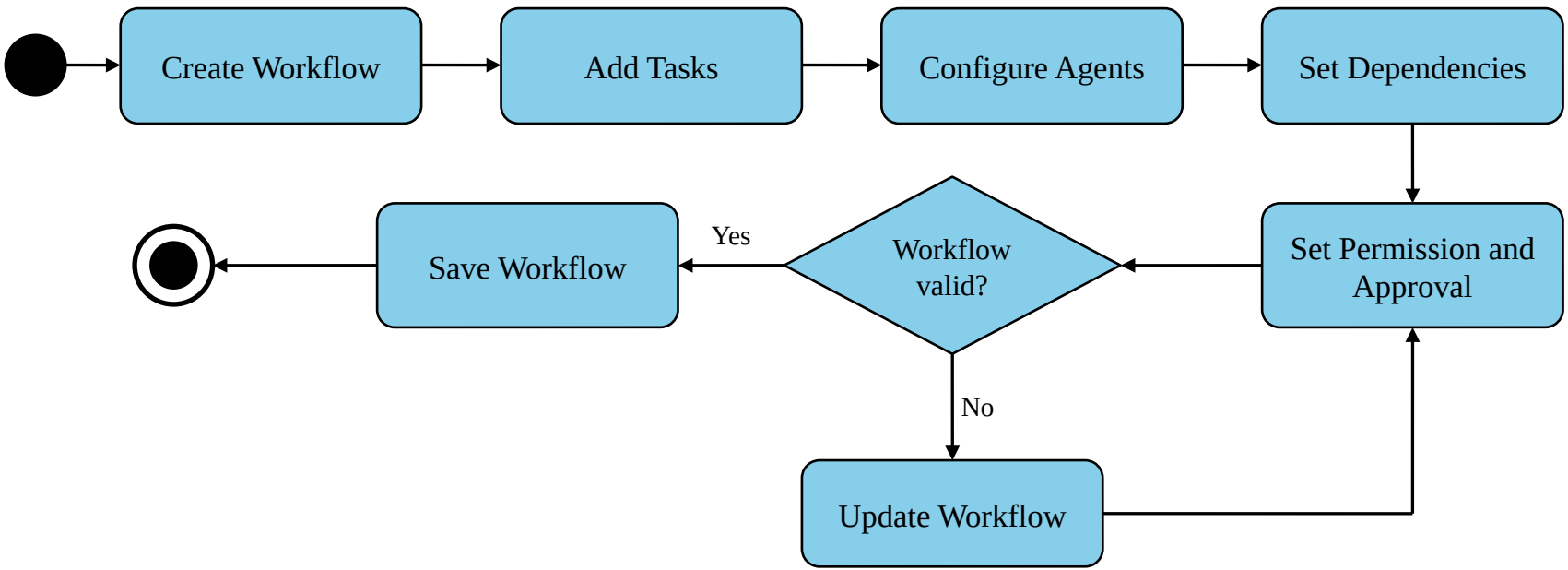
CRC Card: Workflow Orchestrator

Responsibilities	Collaborators
Receives a goal from the user	User
Divides the goal into smaller tasks	Workflow, Task
Assigns tasks to suitable AI agents	Task, AI Agent
Checks the progress of the workflow	Workflow, Task
Handles errors when a task fails	Task, AI Agent

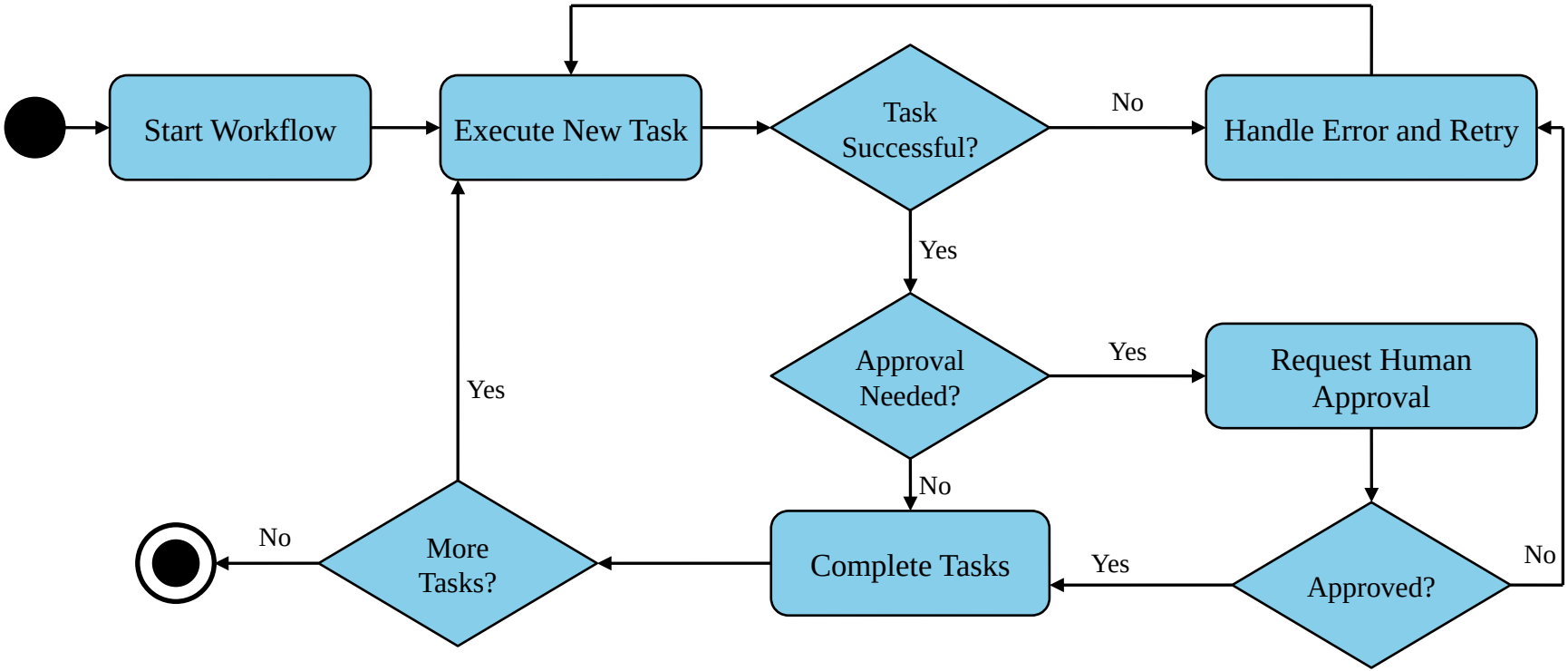
Rough Class Diagram



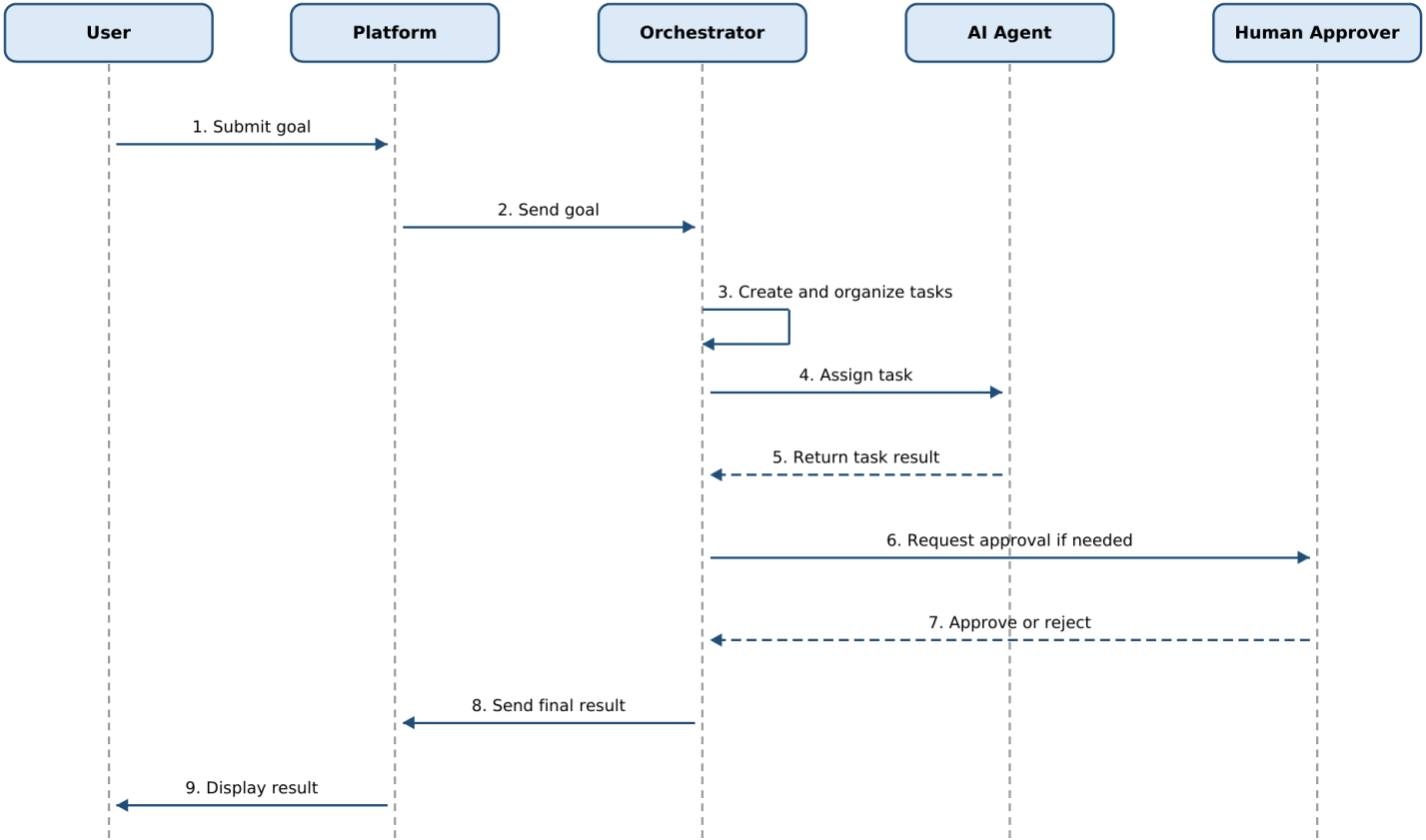
Activity Diagram 1: Create a Workflow



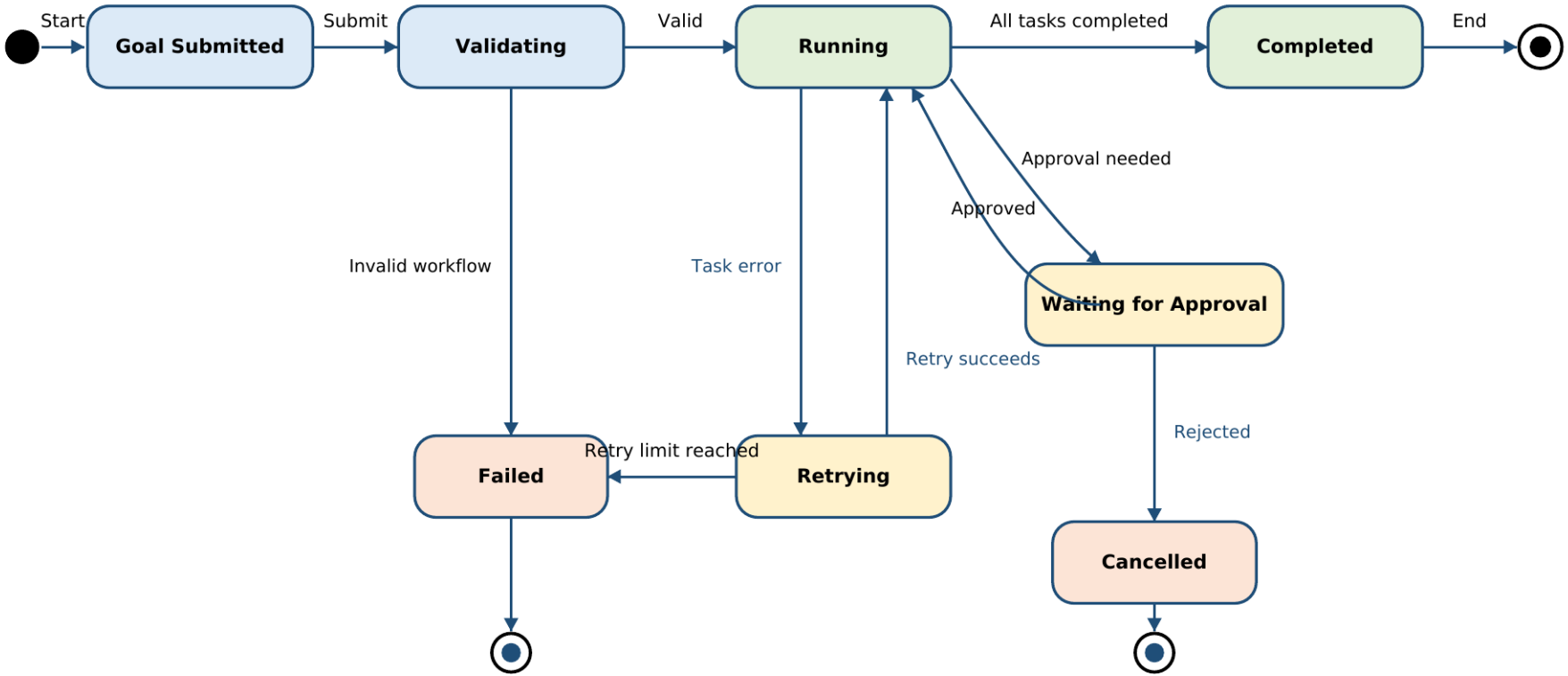
Activity Diagram 2: Execute a Workflow



Sequence Diagram



State Machine Diagram



Timeline and Anticipated Problems

Dates	Milestone
Oct 7 - Oct 13	Requirements, architecture, data model, repo and CI
Oct 14 - Oct 20	Basic orchestrator, sequential execution
Oct 21 - Oct 27	Agent configuration, tools, permissions
Oct 28 - Nov 3	Parallel execution, dependencies, backend API, database
Nov 4 - Nov 10	Workflow designer and basic dashboard
Nov 11 - Nov 17	Error handling, retries, approval gates, monitoring
Nov 18 - Dec 2	Integration testing, demos, bug fixing, evaluation

Problem	Plan
LLM unreliability	Structured outputs, schema validation, reviewer agents, retries
Task decomposition quality	Validate task graphs, allow human edits
Cost and latency	Step and token caps, caching, smaller models
Error propagation	Checkpointing, per-step validation, partial re-runs
Security and safety	Per-agent permission scopes, human approval gates

Thank You !